

Falsifier T1 design v0.1 — lepton-sector COUNT (24 components) from canonical-basis $\dim 4 \times \text{SU}(2)_L \text{ doublet} \times 3 \text{ generations}$ (with Cal, Keeper Saturday plan P4.5 extension).
Forced YES/NO via standard QG computation: $\dim V(1/2,1/2)$ of $U_q(B_2) = 4$ (Dirac 4-spinor); SM observed 24 lepton components matches substrate prediction (Dirac); falsifier triggers on 4th generation, $0\nu\beta\beta$ (Majorana), non-doublet structures, or SUSY partners.

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Falsifier T1 — lepton-sector count from canonical-basis dim

0. The design intent

Keeper requested a sharp QUANTITATIVE falsifier of the keystone bet — beyond P4.5's PMNS angle channel (F1) — that compares the dim of the canonical basis at $V_{-}(1/2,1/2)$ against pinned-a-priori SM counts, computable via standard quantum-group theory with no dictionary-bet leakage in the QG step.

1. The QG-rigorous input (no dictionary leakage)

Canonical basis of $V_{-}(1/2,1/2)$ of $U_q(B_2)$ at $q=2$: - Standard Lusztig-Kashiwara canonical basis on the spinor module. - The spinor module of $U_q(B_2)$ at any q is the 4-dimensional fundamental rep (the Dirac 4-spinor). - $\dim = 4$ (Weyl dim formula: $(a-b+1)(a+b+2)(2a+3)(2b+1)/6$ at $(a,b)=(1/2,1/2)$ gives 4). - The 4 basis elements correspond to the 4 spinor weights $(\pm 1/2, \pm 1/2)$ of B_2 .

This is rigorous representation theory, no dictionary bet involved.

2. The substrate prediction (with dictionary inputs)

The keystone bet says: K-types ARE particles. The dictionary places the lepton at $V_{-}(1/2,1/2)$ (4 components). The physical lepton sector also has: - **SU(2)_L doublet** structure (electroweak): per generation, leptons come as (ν, e) doublet $\rightarrow$ factor of 2. - Justification: L2 derived the Shilov stabilizer $SO(4) = SU(2)_L \times SU(2)_R$; $SU(2)_L$ organizes the doublet. - 3 generations (Route II of #414): per dictionary, generations = $h^\vee = N_c = 3$.

So the substrate-predicted lepton-sector count:

$$N_{\text{lepton}} = (\dim V_{-}(1/2,1/2)) \times (\text{doublet factor}) \times (\text{generation count}) = 4 \times 2 \times 3 = 24 \text{ components}$$

This includes particles + antiparticles + chiralities + generations, with Dirac neutrinos (per F5 prediction).

3. The SM observed count

With Dirac neutrinos (substrate prediction per F5): - Per generation: 1 charged lepton (Dirac, 4 components: $\psi_L, \psi_R, \bar{\psi}_L, \bar{\psi}_R$) + 1 neutrino (Dirac, 4 components) = 8 lepton components. - 3 generations $\times$ 8 = 24 lepton components total.

MATCHES substrate prediction = 24.

With Majorana neutrinos (alternative scenario): - Per generation: charged lepton (4 components) + neutrino (2 components: self-conjugate) = 6 lepton components. - 3 generations $\times$ 6 = 12 lepton components total. - This would NOT match substrate's 24 prediction.

4. The falsifier T1 — explicit YES/NO

Current status: YES (CONFIRMED, modulo neutrino Dirac/Majorana question)

Triggers (falsifier NO):

Observation	Substrate predicted	If observed
4th generation discovered	3 generations $\times$ 8 = 24	$4 \times 8 = 32$ (8 over substrate; CONTRADICTS)
$0\nu\beta\beta$ observed $\rightarrow$ Majorana neutrinos	Dirac, 24 components	Majorana, 12 components (substrate Dirac assumption fails)
Non-doublet lepton multiplet (e.g., triplet)	2-doublet $\rightarrow$ 24	triplet $\rightarrow$ extra components $\rightarrow$ falsifies SU(2) _L organization
SUSY lepton partners	24 components (no SUSY per F5)	48 components (doubled) — refutes F5 + T1
K-type beyond $V_{(1/2,1/2)}$ hosts a charged lepton	dim 4 fits	extra K-type $\rightarrow$ extra components $\rightarrow$ falsifies K-type-as-particle mapping

Any single positive observation in the right column $\rightarrow$ T1 fails $\rightarrow$ keystone bet challenged.

5. CD caveats (Cal #27 honest framing)

The QG dim=4 step is RIGOROUS (Weyl formula, no dictionary). The 24 count multiplies in TWO dictionary outputs: - Doublet factor (2) — from L2 Shilov stabilizer SO(4) $\rightarrow$ SU(2)_L (dictionary structural). - Generation factor (3) — from route II $h^{\wedge}v = N_c = 3$ (dictionary mechanism, currently open with burden).

So: - Cleanest pure-QG falsifier: dim 4 (per K-type $V_{(1/2,1/2)}$) MUST match observed 4-component Dirac structure of ONE lepton (charged or neutrino) — verified [7]. - Combined falsifier: $24 = 4 \times 2 \times 3 =$ total lepton component count — multiplies dictionary outputs; falsification on this channel could indicate either keystone failure OR dictionary-mechanism failure (need to disambiguate).

Per Cal #27: state explicitly which level (per-K-type dim 4 = pure QG; total 24 = dictionary-combined).

6. Honest scope + tier

RIGOROUS (QG, no dictionary bet): - dim $V_{(1/2,1/2)}$ of $U_q(B_2)$ at $q=2 = 4$ (Weyl formula). - The Dirac 4-spinor structure matches this dim.

DICTIONARY-DEPENDENT: - doublet factor 2 from L2 (Shilov stabilizer). - generation count 3 from route II. - 24 total prediction.

FALSIFIER CHANNELS (T1): - Per-K-type (pure QG): dim 4 = observed Dirac-spinor structure [7]. - Total (dictionary-combined): $24 =$ observed lepton components (Dirac scenario) [7]. - NO triggers: 4th generation; $0\nu\beta\beta$; non-doublet; SUSY partners; extra-K-type leptons.

Cal #27 / honesty: T1 is a SHARP quantitative falsifier — but with explicit CD layering (pure-QG core + dictionary-multiplied total). The “24 = observed lepton count” match is genuine but multiplies in dictionary outputs; the cleanest pure-QG falsifier is the per-K-type dim 4 matching Dirac-spinor structure. Both layers MATCH currently; falsifier is LIVE for multiple channels.

Co-design with Cal: this v0.1 frames the test; Cal’s input — pre-write referee questions; identify additional disambiguation steps; sharpen the CD layering — would advance it to v0.2.

Routed: → Cal: cold-read this v0.1 falsifier design; sharpen the dictionary-leakage layering; pre-write referee questions on the YES/NO channels. → Casey: T1 + F5 together (Majorana $0\nu\beta\beta$ + 4th-generation absence) are the cleanest near-term tests of the keystone bet — both have explicit experimental programs (LEGEND-1000, nEXO; HL-LHC). → me: T1 closes the falsifier suite; continuing to EOD prep.

— Lyra, Falsifier T1 v0.1 — lepton-sector count from canonical-basis dim. RIGOROUS (QG): $\dim V_{-}(1/2,1/2) = 4$ (Dirac 4-spinor). DICTIONARY-COMBINED: $24 = 4 \times 2 \text{ doublet} \times 3 \text{ generations}$ matches observed Dirac SM. NO triggers: 4th generation, $0\nu\beta\beta \rightarrow$ Majorana, non-doublet lepton, SUSY partners, extra-K-type leptons. Pure-QG dim 4 matches observed Dirac-spinor structure $\boxtimes$ — cleanest layer. Combined 24-count match also $\boxtimes$ — dictionary-layered. Currently the YES-match holds at both layers; falsifier LIVE.